

A Deliberate Architecture for AI-Assisted Sickle Cell Diagnosis

A Technical Case Study in Building Trustworthy
and Effective Healthcare AI

The Clinical Challenge Demanded a Multi-Modal AI Solution

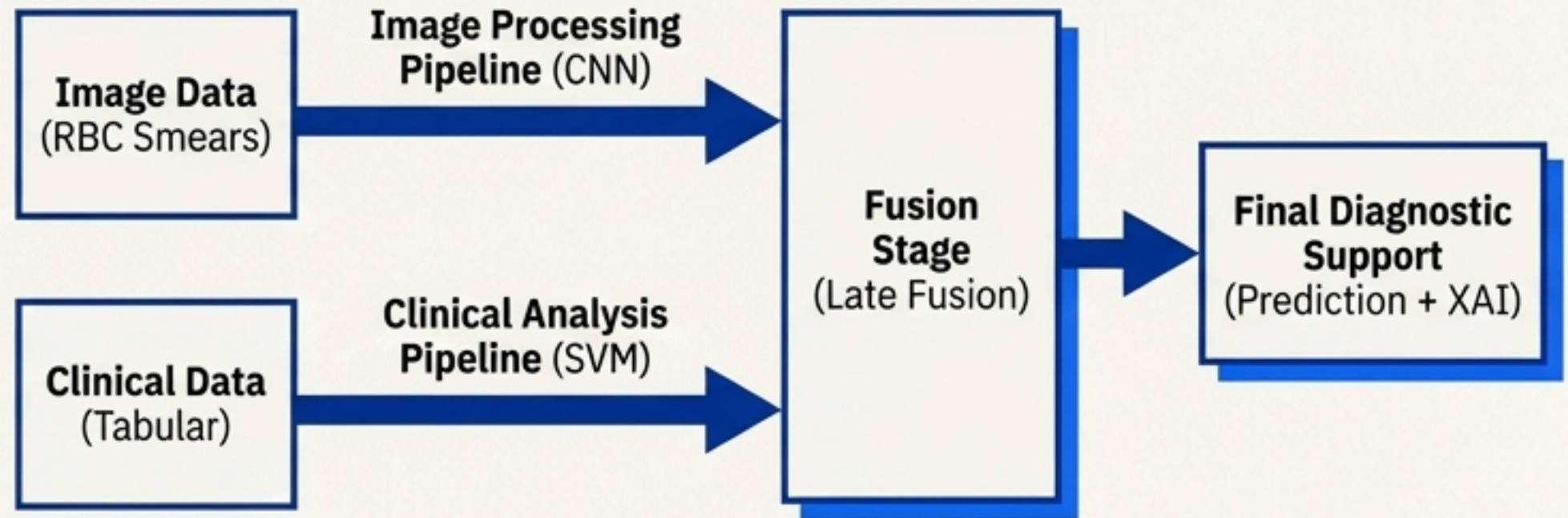
The Problem

Sickle cell diagnosis traditionally relies on manual microscopic analysis of Red Blood Cell (RBC) images and separate clinical tests. This process is:

- 🕒 • **Time-consuming & Subjective:** Prone to human variability.
- 🔄 • **Difficult to Scale:** A bottleneck in resource-constrained settings.

The Architectural Response

To mirror the real-world diagnostic workflow, we designed a **dual-pipeline framework** that analyses both image and clinical data independently before fusing their predictions.



High-Fidelity Results Guided by Core Engineering Principles

99.1%

Accuracy on RBC Image Classification

96%

Accuracy on Clinical Data Analysis

Core Principles



1. Clinical Realism

The architecture mirrors how clinicians work with multi-modal data, improving robustness.



2. Pragmatic Model Selection

Every model and library was chosen as the 'best tool for the job,' not forced into a single framework.



3. Trust through Transparency

Explainable AI (XAI) was a mandatory requirement from day one to build clinician confidence and ensure safety.

A Dual-Pipeline Architecture Was Chosen to Isolate and Optimise Heterogeneous Data

Challenge

Image data (high-dimensional, spatial) and clinical data (low-dimensional, tabular) have fundamentally different statistical properties. Fusing them too early (early fusion) can lead to optimisation conflicts and negative transfer.

Decision

Implement two independent, complementary processing pipelines that operate in parallel and are combined only at the final prediction stage (late fusion).

Rationale

- **Independent Optimisation:** Allows for the use of the most suitable model for each data type (CNN for images, classical ML for tabular).
- **Improved Robustness:** The system can still provide a prediction if one modality is noisy or unavailable.
- **Enhanced Interpretability:** Decoupling the pipelines makes it easier to debug and understand feature contributions from each source.

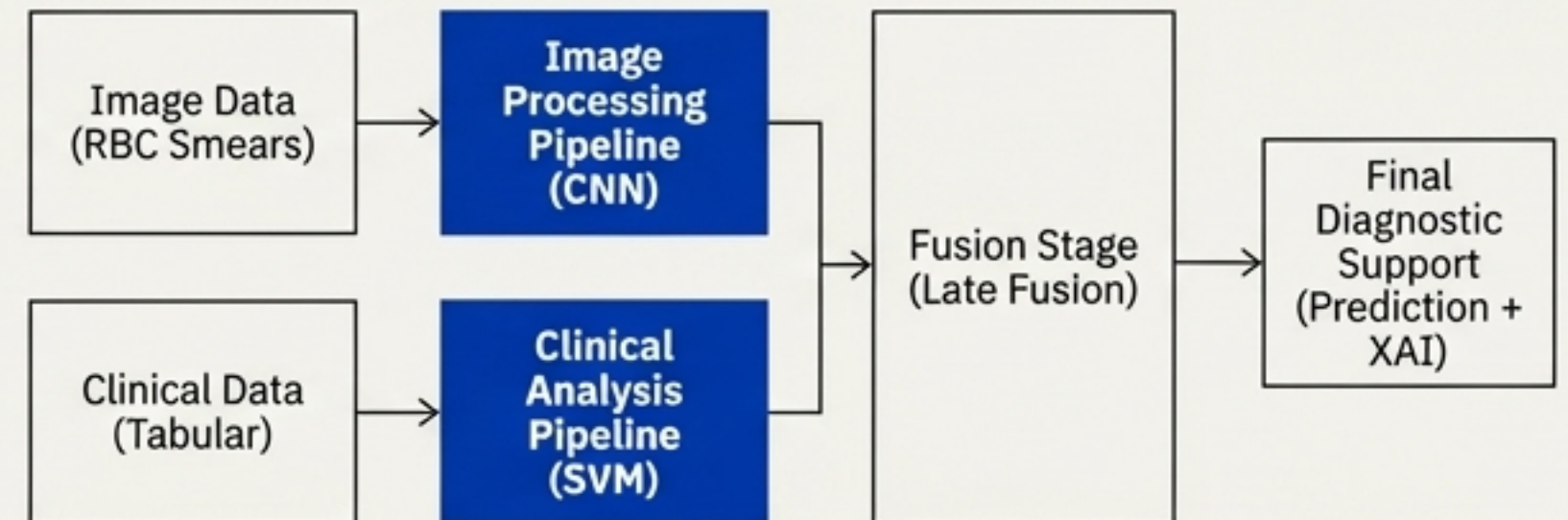


Image Pipeline: EfficientNet & Transfer Learning

Mitigate Small Dataset Constraints

Challenge

Medical image datasets are typically limited in size, making deep learning models trained from scratch prone to overfitting and computationally expensive.

Decision

Use EfficientNet-B0, a lightweight pre-trained CNN, and apply transfer learning.

```
# Load pre-trained model without the  
final classification layer  
base_model = EfficientNetB0(  
    weights='imagenet',  
    include_top=False,  
    input_shape=(224, 224, 3)  
)
```

Rationale

- **Leverages Pre-trained Features:** Utilises robust low-level features (edges, textures) learned from ImageNet, which are universal and relevant to RBC morphology. This drastically reduces data requirements.
- **Superior Efficiency:** EfficientNet's compound scaling (balancing depth, width, resolution) achieves **higher accuracy** with fewer parameters and lower computational cost compared to older architectures.
- **Better Generalisation:** Avoids the 'inefficient and risky' proposition of training a large custom CNN that would likely memorise the small dataset.

Clinical Pipeline: SVM Was Selected for Stability on Low-Dimensional Data

Challenge

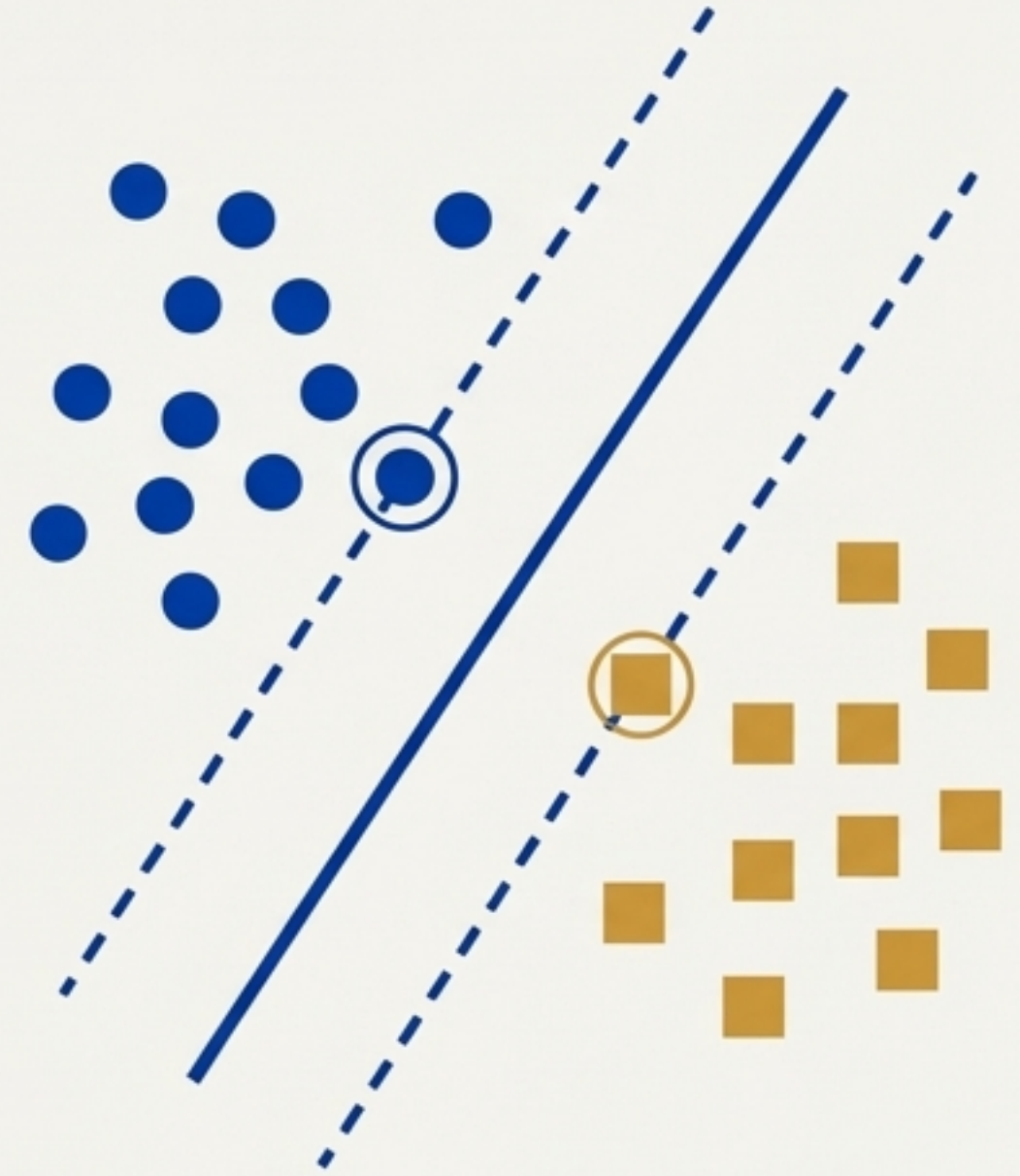
Clinical data is typically tabular, low-dimensional, and can be noisy. Deep learning models often require large datasets to perform well and can add unnecessary complexity.

Decision

Employ a Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel.

Rationale

- **Performs Exceptionally on Small Datasets:** SVM's principle of margin maximisation provides strong generalisation guarantees, even with limited data.
- **Handles Non-Linearity:** The RBF kernel effectively captures complex, non-linear relationships between clinical features.
- **More Stable & Interpretable:** Compared to deep neural networks for tabular data, SVMs are more stable and less of a 'black box,' making them a safer choice in a clinical context.



Data Integrity: SMOTE Was Deliberately Chosen to Address Clinical Class Imbalance

Challenge

In medical datasets, the minority class (e.g., sickle cell patients) is often the most clinically critical. A naive model can achieve high accuracy by simply ignoring it, which is dangerous.

Decision

Apply the Synthetic Minority Over-sampling Technique (SMOTE) *only to the training data*.

Rationale - A Comparison of Techniques



✗ **Why Not Random Oversampling?** Creates duplicate data, leading to overfitting.

✗ **Why Not Undersampling?** Discards valuable clinical information, risky in small datasets.

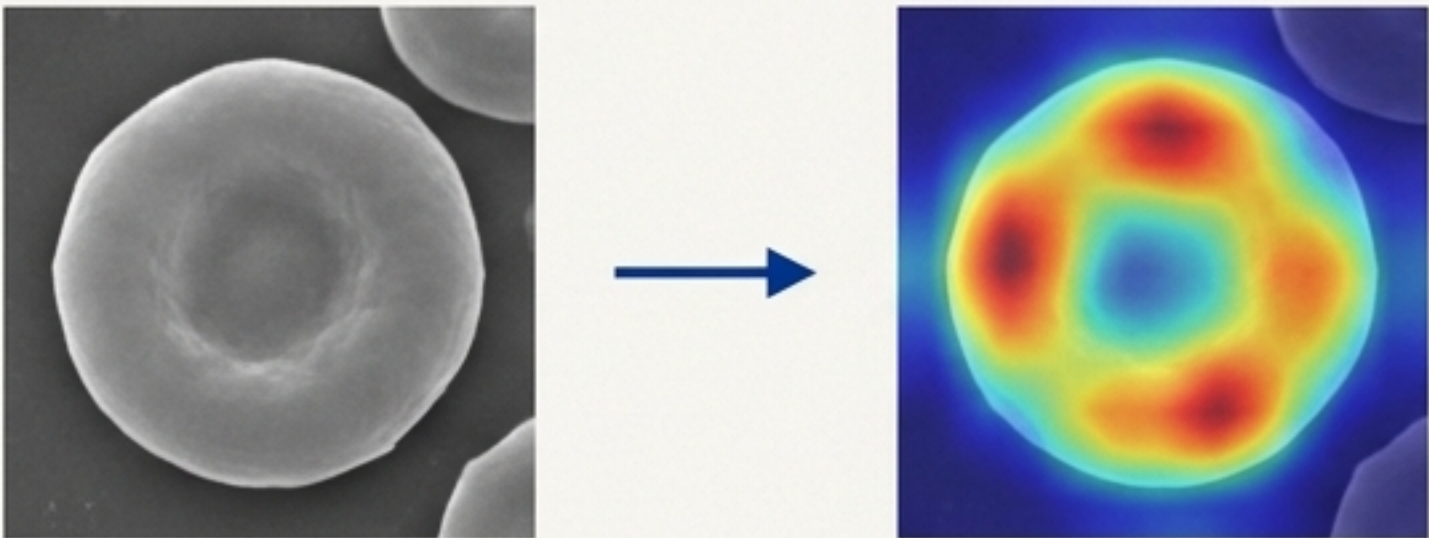
✓ **Why SMOTE Was Best:** It generates new, realistic minority samples by interpolating between existing ones. This balances the dataset without losing information or creating simple copies, preserving feature relationships and decision boundaries. This is clinically safer than generating samples in noisy regions (unlike ADASYN).

The Cornerstone of Adoption: Building Clinician Trust with Explainable AI

In healthcare, correctness alone is insufficient—trust, transparency, and usability are equally critical. Accuracy tells us *what* the model decided; XAI tells us *why*.

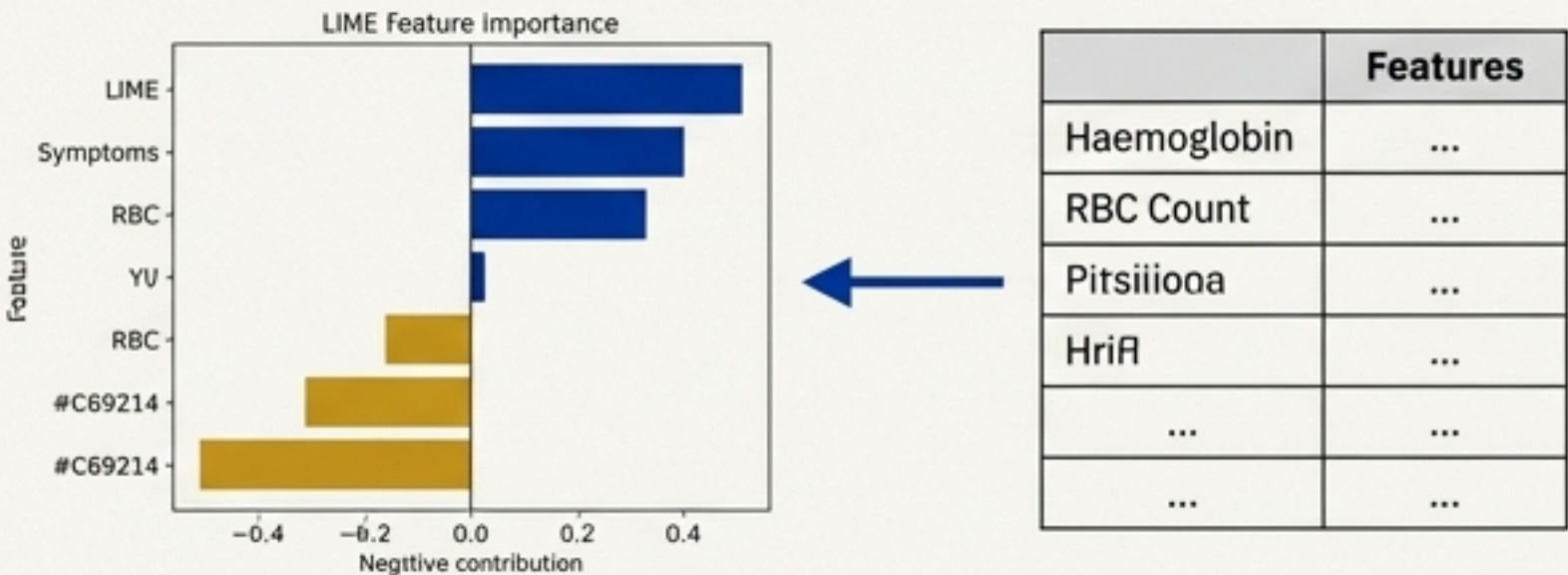
Our Dual-XAI Approach: We implemented a specific explainability method for each data modality to provide clinicians with clear, actionable insights.

For Image Diagnosis



Grad-CAM visualises which parts of an RBC image influenced the model's decision.

For Clinical Diagnosis



LIME identifies which patient data points (e.g., haemoglobin levels, symptoms) contributed most to the prediction.

Visual Explainability: Grad-CAM Aligns with CNNs to Create Intuitive Heatmaps

Challenge

Provide a visual explanation that is intuitive for clinicians and directly maps to the input image's morphology.

Decision

Integrate Grad-CAM (Gradient-weighted Class Activation Mapping).

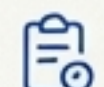
Rationale



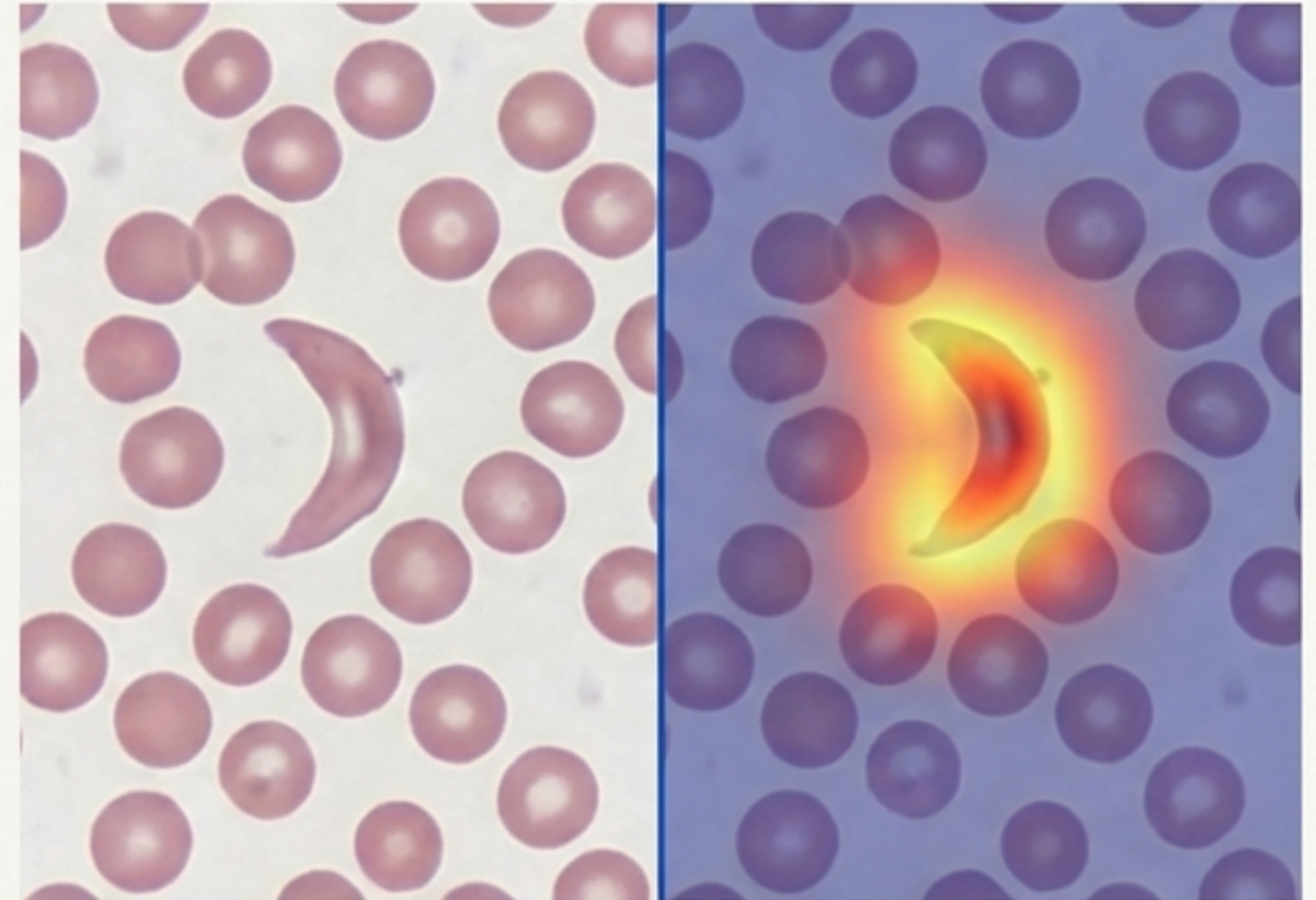
Clinically Intuitive: Produces visual heatmaps that highlight the exact regions of an image the model focused on, allowing experts to verify if the reasoning is sound (e.g., focusing on sickle-shaped cells).



Natural Synergy with EfficientNet: EfficientNet has deep, well-structured convolutional blocks whose final layers retain rich spatial information. Grad-CAM leverages these layers to produce clean, stable, and meaningful activation maps.



Established and Trusted: It is a widely used, peer-reviewed method in medical AI research, adding to the system's credibility.



The Grad-CAM heatmap (right) precisely highlights the sickle cell, providing a powerful, self-evident demonstration of the model's focus on clinically relevant morphology.

Clinical Explainability: LIME Was Prioritised Over SHAP for Real-Time Interaction

Challenge

For the clinical data pipeline, provide fast, patient-specific explanations of what features are driving the SVM’s prediction in an interactive user interface.

Decision

Use LIME (Local Interpretable Model-agnostic Explanations).

Rationale: A Pragmatic Trade-off

	Criteria	LIME	SHAP
1	Real-time UI	Fast	Slower (computationally expensive for kernel SVM)
2	Explanation Type	Strongly Local (patient-specific)	Mixed (can be global)
3	Simplicity for Doctors	Clear, simple feature weights	Can produce dense, harder-to-interpret plots

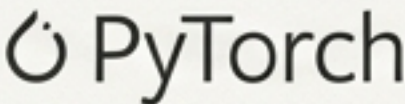
Conclusion: While SHAP is more theoretically grounded, LIME provided the best balance of speed and interpretability for our primary requirement: providing a clinician with a fast, easy-to-understand explanation for a single patient’s prediction in real time.

A Pragmatic Tech Stack: Selecting the Best Tool for Each Component's Job

The Philosophy

Instead of forcing all logic into a single, monolithic framework, we selected each library for its specific strengths. This mirrors real-world ML systems where flexibility and performance matter more than uniformity.

System Component Breakdown

	Scikit-learn - The industry standard for structured data; stable, well-optimised, and integrates easily with LIME.
	TensorFlow / Keras - Excellent support for EfficientNet, simple and stable transfer learning workflow.
 PyTorch	PyTorch - Offers finer-grained control over gradient manipulation, making it ideal for experimental and flexible Grad-CAM implementations.
 Streamlit	Streamlit - Rapidly translates Python scripts into interactive web applications for model validation.

Deployment: Streamlit Was Used for Rapid Prototyping and Clinician Feedback

Challenge -> Decision -> Rationale

Challenge

The primary goal was the rapid, interactive deployment of an AI diagnostic tool for clinicians, not full-scale web engineering.

Decision

Build the user interface with Streamlit.

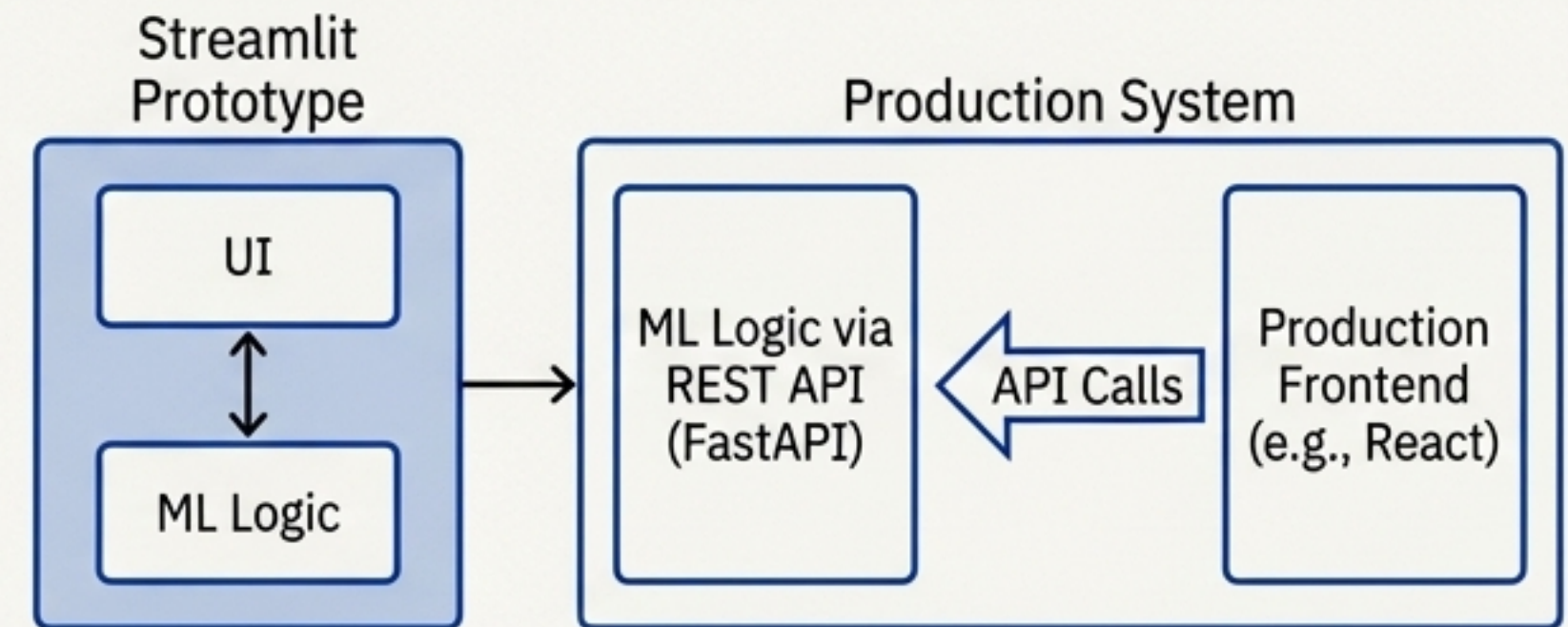
Rationale

- **Speed of Development:** Enabled extremely fast prototyping, turning Python ML scripts directly into a shareable web app.
- **Focus on AI, Not Plumbing:** The project's value is in the AI logic and explainability, not in backend complexity like routing or authentication.
- **Seamless ML Integration:** Directly compatible with all the ML frameworks used.

The Path to Production

Streamlit was ideal for this research-grade prototype.

For a production-scale hospital deployment, the system would evolve: the ML logic would be exposed via a robust REST API (using FastAPI), and a separate, production-grade frontend would be developed.



Engineering Maturity: Understanding When and Why to Evolve the Architecture



Guiding Principle

The selected models are not universally optimal; they are the best fit for the current project's specific constraints (data size, compute, explainability needs).

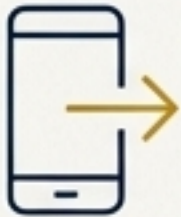
Future Evolution Scenarios

When would you replace EfficientNet?



If Data Grew 10x

Consider Vision Transformers (ViT) or larger EfficientNet variants (B3/B4) to capture more global context.



If Deployed to Mobile/Edge

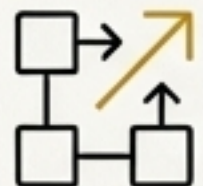
Prioritise latency and power efficiency with MobileNetV3 or Quantized CNNs.

When would you replace SVM?



If Clinical Dataset Became Much Larger

Explore Gradient Boosting models (like XGBoost) or Deep Tabular models for their ability to leverage more data.



Conclusion

The architecture is designed to be modular, allowing for these components to be replaced as data, hardware, and requirements evolve.

A Proactive Approach to Risk Mitigation, Safety, and Ethical AI

Key Risks and Mitigations

Model Overfitting & Performance Traps

- 🛡️ **Mitigation.** Used transfer learning, dropout, and a strictly separated test set. Validated performance using recall and precision, not just accuracy, to minimise the risk of false negatives.

Data Leakage During Preprocessing

- 🛡️ **Mitigation.** SMOTE and feature scaling were applied *only* after the train-test split, ensuring the model was never exposed to test data information.

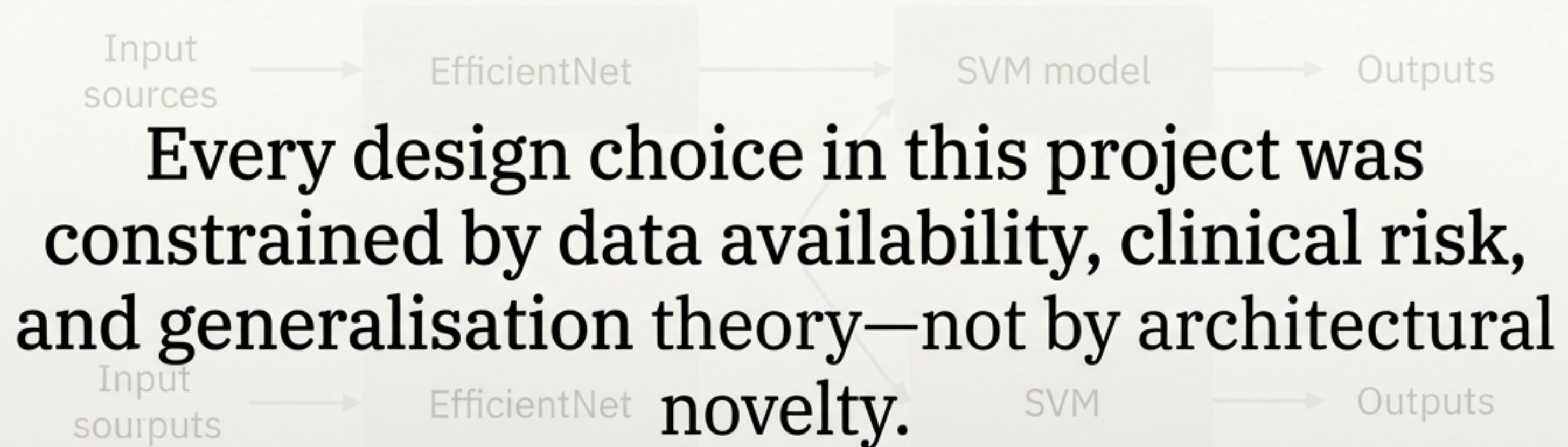
Misinterpretation of 'Black Box' AI

- 🛡️ **Mitigation.** Integrated XAI (Grad-CAM, LIME) from the start. The system is designed as a decision-support tool to augment, not replace, clinical judgement.

Unrealistic XAI Explanations

- 🛡️ **Mitigation.** Validated that Grad-CAM explanations aligned with known medical pathology. Acknowledged that XAI provides approximations that require expert verification.

Final Principle: A System Designed for Trust, Not Just Accuracy



The final system prioritises patient safety, transparency, and validation over raw performance, because in healthcare, trust is as critical as accuracy.